

Problem 6 - Stochastic Gradient Descent

i) $P(\text{not choosing a sample}) = \frac{n-1}{n} = 1 - \frac{1}{n}$
 with replacement $\Rightarrow n$ independent events

$$\Rightarrow \left(1 - \frac{1}{n}\right) \times \dots \times \left(1 - \frac{1}{n}\right)$$

$$\Rightarrow \boxed{P_{\text{never}} = \left(1 - \frac{1}{n}\right)^n}$$

ii) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$; Show that $\lim_{n \rightarrow \infty} P_{\text{never}} \approx 0.3674$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n$$

$$\text{let } y = \left(1 - \frac{1}{n}\right)^n$$

$$\ln(y) = n \ln\left(1 - \frac{1}{n}\right)$$

$$\lim_{n \rightarrow \infty} \ln(y) = \lim_{n \rightarrow \infty} \frac{\ln\left(1 - \frac{1}{n}\right)}{1/n} = \frac{\left(-\frac{1}{1-n}\right)\left(-\frac{1}{n^2}\right)}{-\frac{1}{n^2}} = -\frac{1}{1-n} = -1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \ln(y) = -1$$

$$\Rightarrow \boxed{\lim_{n \rightarrow \infty} y = e^{-1} = 1/e \approx 0.3679}$$

iii) One random observation $\text{Var}(\nabla f(x^{(i)}, y^{(i)})) = \sigma^2$
 Sampling with replacement $\Rightarrow$ random variables are i.i.d
 for k iid random variables with variance σ^2

$$\text{sample mean } \bar{X} = \frac{1}{k} \sum_{i=1}^k X_i$$

$$\boxed{\text{Var}(\bar{X}) = \frac{\sigma^2}{k}}$$

iv) The variance is reduced by a factor of k

v) Since increasing the batch size by k leads to a reduction in variance, increasing the learning rate may introduce more noise. You may want to reduce the learning rate to keep the noise constant.